Graph-Fourier Power-Flow Operators and the Feasibility Gap in Learned Grid Control

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Abstract

A learned controller for a transmission network has to satisfy thermal and voltage limits it cannot see, and the usual remedy is a learned power-flow surrogate in front of it. We build that surrogate on the graph Fourier basis of the network's own Laplacian, so its global mixing channel is indexed by frequency rather than by position, and we wrap it in a projection shield that scores every admissible topology option in one batched operator pass and repairs an unsafe action before it is executed. Instrumenting the pair shows that the basis choice is not where the safety gap comes from. Against a Newton-Raphson solver verified to 9.379×10^{-9} p.u. against pandapower over 59 states, a contingency moves the eigenvalues by 0.007375 of the largest one but rotates the eigenvectors until their mean overlap with the intact modes is 0.412, so a frequency-indexed filter is evaluated where it was trained and still multiplies different content. A rasterised Fourier operator accordingly costs $1.038\times$ the held-out error of the frequency-indexed one at $3.221\times$the parameters, while a message-passing network that never commits to a basis a contingency can invalidate is the most accurate of the six, and a surrogate that costs more than the solver at a single query pays for itself only in batches. At these accuracies every held-out state is called infeasible, so the shield's contribution is structural, removing 0.8966 of the islanding events the unshielded controller causes at 24.5 exact solves per interval, and its residual is a 0.8733 violation rate rather than a guarantee, which leaves the verdict rather than the return as the quantity a safety claim has to be read from.

1 Introduction

Operators of transmission networks know how to compute a secure dispatch by solving a non-convex alternating-current optimal power flow, which they cannot do inside the five-minute interval in which a wind farm drops or a data centre ramps. The field reaches for learned surrogates because such a map is evaluated once and reused while the classical solve is paid for on every query, and the obstruction is not average accuracy, since a surrogate can be accurate on average and wrong at the limit that matters.

A second obstruction sits underneath the first, since a transmission network is not a lattice, its structure changes, and choosing that change is part of the operator's decision. A standard Fourier neural operator mixes in the discrete Fourier basis of a regular grid, so on a network that basis must be manufactured by embedding buses at coordinates and rasterising, and a contingency then leaves every learned spectral weight evaluated somewhere other than where it was learned. This paper keeps the architecture and replaces the object it acts on, using the eigenbasis of the network's own Laplacian Shuman et al. [2013], Li et al. [2020]. Two quantities decide what that buys and they have opposite sizes, since across 38 single-line contingencies the eigenvalues move by only 0.007375 of the largest one on average, which leaves a frequency-indexed filter evaluated close to where it was trained, while the eigenvectors rotate until their mean overlap with the intact modes is 0.412 and the

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worst retained mode overlaps by 0.03835. The frequency-assignment problem this paper set out to fix is real but small, and the fixed-basis problem underneath it is large. The shield we build on that basis scores every admissible topology option in one batch, refuses any switch that islands the network, and descends on the predicted constraint violation with respect to the redispatch, which is the only place differentiability is required. It never calls the exact solver, because such a shield would inherit the runtime that motivates the surrogate, and measured end to end it removes 0.8966 of the islanding events the unshielded controller causes while approving nothing, so it reduces to a projector whose residual is a violation rate rather than a false-safe rate.

2 Problem formulation

A network is a connected graph with n_b buses and n_ℓ branches. Generator g sits at bus $\beta(g)$ with active output P_g, a voltage setpoint V_g^* and limits $P_g^{\min} \le P_g \le P_g^{\max}$. Branch ℓ joins buses f_ℓ and t_ℓ with series impedance $r_\ell + \mathrm{j}x_\ell$, shunt susceptance b_ℓ and thermal rating R_ℓ. The state is the complex bus voltage $V_i = V_{m,i}\mathrm{e}^{\mathrm{j}\theta_i}$, and $S(V) = \mathrm{diag}(V)\overline{YV}$ relates it to the injections. A branch's loading $\rho_\ell = \max(|S_{f,\ell}|, |S_{t,\ell}|)/R_\ell$, the larger of its from- and to-side apparent powers over its rating, defines the thermal constraint $\rho_\ell \le 1$, and $V_i^{\min} \le V_{m,i} \le V_i^{\max}$ is the voltage constraint. A topology is the in-service mask $\sigma \in \{0,1\}^{n_\ell}$, and an islanded state is undefined rather than infeasible, so a disconnection is charged as a blackout and never as a constraint violation. An episode is T intervals in which the controller chooses a redispatch $\Delta P \in \mathbb{R}^{n_g}$ on the non-slack generators together with at most one branch switch, minimising operating cost subject to the constraints holding in every interval, a constrained Markov decision process whose cost and constraint channels stay separate so that safety is read off at evaluation. Every physics number comes from a dense polar Newton-Raphson solve, verified against pandapower on 59 states to 9.379×10^{-9} p.u. in voltage magnitude, 6.423×10^{-7} degrees in angle and 1.217×10^{-8} p.u. in branch flow (Appendix A).

3 Method

The graph Fourier basis of a topology is the eigenbasis of its symmetric normalized Laplacian $L = I - D^{-1/2}AD^{-1/2}$, with edge weights $w_\ell = \sigma_\ell / x_\ell$. The weight is series susceptance, the linearised angle coupling of the power flow, so L is the direct-current operator on the in-service network. With U the eigenvector matrix and λ the eigenvalues in ascending order, a spectral block computes

$$y = U\left[\, W(\lambda) \odot \mathrm{mask}(\lambda) \odot (U^\top x) \,\right], \qquad W(\lambda) = \textstyle\sum_j B_j(\lambda)\,\Theta_j, \tag{1}$$

where x is the latent at the nodes, B_j are cubic B-spline basis functions over $[0, \lambda_{\max}]$, Θ_j are learnable and the mask is the retention rule. The two rules we compare are $\mathrm{mask}_{\mathrm{idx}} = \mathbf{1}\{k < K\}$ applied to the sorted index k, which is what a standard Fourier neural operator does on a lattice, and $\mathrm{mask}_{\mathrm{band}} = \mathbf{1}\{\lambda \le \alpha\lambda_{\max}\}$ with the weight read off at each retained mode's own eigenvalue. Both variants carry the same spectral slots, channels and edge paths, so a difference between them is the retention rule (Appendix C).

The eigenvalues move under a contingency, so an index-indexed weight is evaluated at a frequency it was not trained for, since the per-mode drift $\Delta\lambda_k/\lambda_{\max}$ averages 0.007375 with a worst single-mode value of 0.08544, and $\lambda_{\max}$ itself moves by 0.006749. The eigenvectors also rotate, so a filter evaluated at exactly the right frequency still multiplies different content, because the overlap between intact and contingent eigenvectors averages 0.412 at each retained mode with the worst retained mode at 0.03835. Eigenvalue parameterisation repairs the first problem and does nothing about the second, so the prediction is a small index-versus-band difference and a larger penalty for naming a fixed basis at all (Appendix D). The shield then refuses any switch that islands the network, keeps the predicted-feasible topology option with the lowest operating cost, minimises the predicted constraint violation with respect to the redispatch by gradient descent with each step bounded by that unit's ramping band, and accepts a step only if it does not worsen the worst predicted margin, which makes the projection monotone in that margin, the one guarantee it offers. It executes the result without checking it, so the residual is a measured rate and not a bound, and both inner loops are a single batched pass, which is what keeps a shield that never calls the solver affordable. The controller it wraps is proximal policy optimisation with a categorical head over the switches and a Gaussian head over the redispatch, trained with a Lagrangian multiplier on the constraint channel set to half the

Table 1: Power-flow operator accuracy on a topology-disjoint split of the scenario ensemble. Both columns report per-sample relative L^2 on branch loading against the verified Newton-Raphson solver; the held-out column uses contingencies no model saw in training. Means over 2 seeds, 132 held-out scenarios. The two graph-Fourier retention rules are parameter-matched by construction, since they share the spline coefficients and differ only in the mask over retained modes. Lower is better.

Architecture	Seen flow	Held-out flow	Held-out V_m	Held-out V_a	Params
MLP (no graph)	0.3064	0.3349	0.008959	0.3299	33,741
GNN	0.1867	0.2099	0.006107	0.2398	91,981
FNO, rasterised	0.1784	0.2483	0.01024	0.3546	981,581
Chebyshev NO	0.1807	0.2119	0.005625	0.2283	78,541
Graph FNO, index	0.2042	0.2846	0.01041	0.4015	304,781
Graph FNO, band	0.1955	0.2391	0.009915	0.3537	304,781

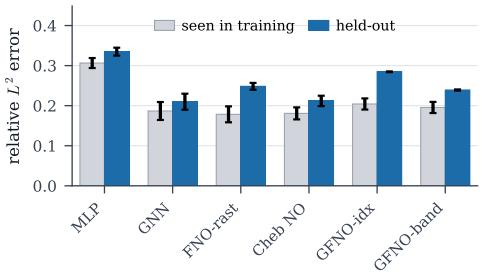


Figure 1: Branch-loading error of the six operators, against the verified Newton-Raphson solver, under a split sharing no contingency between training and testing ($n = 132$ held-out scenarios over $n = 2$ seeds; bars are the mean, segments the spread across seeds; labels are the architectures of Table 1 in the same order, which carries the parameter counts). The rasterised Fourier operator opens the largest gap relative to its capacity and the message-passing network that keeps no spectral basis is the most accurate of the six, so lower is better.

do-nothing violation rate; Appendix C gives its hyperparameters and its observation, which includes the predicted margins.

4 Results

Setup. The network is case30. Ratings are the file's own where they are physical, and where a network carries the format's degenerate rating we substitute 1.25× the peak flow observed on the intact network, then scale all ratings so the intact base case peaks at 0.85 of the tightest branch's limit. The ensemble crosses three intermittency axes, a uniform load multiplier, a data-centre load block on the largest-load buses and a multiplicative cut to the highest-capacity renewable hosts, with contingencies split by rank so that none seen in training appears in testing (Appendix B). The regime is what an $N - 1$ study looks like, with 0.8788 of held-out scenarios violating at least one constraint under the do-nothing dispatch (1.015 thermal and 1.977 voltage constraints per scenario on average); the operator is trained on 129 scenarios and evaluated on 144 and 132.

Result 1: the fixed basis costs more than the frequency assignment. Table 1 and Figure 1 report 6 architectures on identical inputs, budgets and splits, over 2 seeds, and the ordering of the three facts is not the one this framing expected.

First, a filter that is a function of the eigenvalue survives a topology change better than the same filter indexed by mode position, since the band rule loses 1.223× of its accuracy moving from seen to held-out contingencies, from 0.1955 to 0.2391 relative L^2 on branch loading, while the index rule loses 1.393× and ends at 0.2846, a 1.19× difference. Second, and larger, any architecture that must name a fixed basis pays for it, since the rasterised operator places buses at coordinates derived from the current topology before convolving them and a contingency then moves the field underneath the convolution, which leaves it at 0.2483, 1.038× the band rule's held-out error at 3.221× the parameters (981,581 against 304,781), while its seen error is the best of the six, 0.1784 against 0.1955. Third, and against the grain of the whole construction, the model that keeps no spectral basis at all is the best of the six, since a plain message-passing network reaches 0.2099 with the smallest degradation of any learned variant while the topology-blind floor is far behind at 0.3349. Structure matters and

the spectral structure does not pay for itself, because the basis is built from the same Laplacian a message-passing network uses, so the eigenvector rotation measured at 0.412 leaves the filter weight correct while the function it multiplies is wrong.

Result 2: the shield is decisive about structure and blind about precision. Table 3 in Appendix E reports the safety study, judged throughout by the Newton-Raphson solver and never by the surrogate, and the same appendix plots where its verdict is wrong.

Every architecture in Table 1 predicts infeasibility for every held-out state, a rate of 1 against a true 0.8788, so the agreement rate 0.8788 equals the solver's own do-nothing rate 0.8788 and the gap between them is 0, which means a constant verdict scores as well as a discriminating one where most states really are unsafe. The error is one-sided, since 0.1212 of genuinely satisfying states are called unsafe while the false-safe rate is 0,and recalibrating the threshold would not rescue the verdict, since at a five percent false-safe budget the best achievable detection rate is 0.Two arms differ only in whether the shield sits between the policy and the network. The unshielded controller makes the state worse than the do-nothing dispatch in 0.05785 of intervals and better in 0.09091, and attaching the shield moves those to 0.102 and 0.1565, which raises both rates rather than reducing them, because a projected action is a larger change than the policy proposed. What it changes is the net effect, -0.04762 violations per interval against -0.04132 without it, and the failure mode it removes, since refusing any switch that splits the network takes the islanding rate from 0.1933 to 0.02, a 9.667$\times$ reduction. The residual is the counterweight, since the shield never approves an interval unmodified (0) and projects every one (1), so there is no false-safe rate to report and the number a deployment decision would weigh is the rate at which projected actions still violate something, 0.8733 against 0.9133 for the unshielded controller and 0.8788 for doing nothing at all. That move is small, and we attribute its size to the regime, since at the operating boundary most violations are voltage excursions no redispatch inside a ramping band can repair; Appendix E reports the per-arm rates.

Latency: a scope statement, not a claim of speed. One exact solve costs 4.091 ms and one operator forward pass costs 9.244 ms, so at a single query the surrogate is slower, and the comparison turns only when queries are batched, at 5.917$\times$ and 7.461$\times$ faster for batches of 32 and 128 cases. The shield's own runtime follows the same rule, an interval costing it 100.2 ms, or 24.5 exact solves (Appendix D).

5 Discussion and limitations

The contribution is a construction with a measurement attached. We build the operator so that the graph Fourier basis of the network's own Laplacian gives a global mixing channel indexed by frequency instead of position, and wrap it in a projection shield that is decisive about structural failures, and the measurement then shows that the residual safety gap tracks the surrogate's held-out accuracy rather than the policy's, which is why a safety claim has to be read from the verdict rather than the return.

Limitations. The limits are exact, and the projection is monotone in a predicted margin rather than a true one, so it repairs the surrogate's belief and certifies nothing about the grid, which means every safety number here is a measured rate and none is a guarantee. The eigenvalue drift the index-versus-band argument rests on is small on this network, 0.007375 on average, so the difference between the two retention rules is correspondingly small and the stronger form of our own claim is not supported. Islanding is handled by refusal, which makes that reduction a statement about the action space rather than the surrogate's accuracy, and at 24.5 exact solves per interval the shield is justified by what it prevents rather than what it saves. The network is a case30 system with 129 training scenarios, so the latency claim and the safety rates are statements about networks of this size, and testing the mechanism where the spectrum moves further is left to future work.

Broader impact. The quantity that decides how much of a network's capacity an operator can use is whether a controller can be trusted at the limit, and the pathway this work serves is concrete, since screening more switching options inside one operating interval is what a batched surrogate makes possible and refusing the islanding options is what makes the screened set safe to act on. A measured violation rate on a small network is nonetheless not authorization, since a utility would need it to hold under its own contingency set with a margin over the surrogate's worst-case error.

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A Verification protocol for the physics oracle

What is compared. For each network we compare three state families: the base case, every single in-service branch tripped in turn, and every double outage among the five most heavily loaded branches. For each state both solvers are run from a flat start with reactive-limit switching disabled, so the two are solving the same equality-constrained problem rather than two neighbouring ones. The reported quantities are the largest disagreement in voltage magnitude, in voltage angle in degrees, and in from-side branch apparent power in per unit.

Matching pandapower's default. pandapower's `runpp` leaves reactive limits unenforced by default, so it solves the equality-constrained power flow. Our solver has a PV-to-PQ switching path for over-excursion, and running it while pandapower does not produces a disagreement that is a modelling choice rather than an error. The two agree at round-off when the choice is matched, and the switching path is exercised separately by the unit tests.

Islands. A state whose in-service graph is disconnected has no solution, because the component without a slack bus has no reference angle and no source. pandapower reports NaN voltages there and our solver reports a diverged Newton iteration, so the two agree on the verdict rather than on a number. Those states are counted and reported separately, which is also the labelling the environment uses to charge a blackout.

Unit conventions. `loading` in our solver is the branch's apparent-power flow in MVA over its MATPOWER rating in MVA. The flow returned by the admittance product is in per unit, and omitting the system base rescales every thermal limit by a factor of 100 and makes a safety study measure nothing, so the conversion is asserted by a unit test.

B Scenario ensemble and its calibration

Intervals. An episode runs for a fixed number of control intervals. At each interval the load multiplier and the renewable multiplier random-walk with per-interval innovations, so consecutive intervals are correlated the way a real five-minute sequence is and a policy cannot solve the task by re-identifying an independent draw. A new branch trips with a fixed small probability and a tripped branch returns with a smaller one.

Ratings and the operating point. MATPOWER's files for several standard networks carry a single degenerate rating equivalent to declaring every branch unconstrained. Where a rating is more than two orders of magnitude above anything the branch can carry we substitute 1.25 times that branch's peak flow observed on the intact network, taken on the intact topology rather than over the contingency pool, because a rating derived from a branch's own worst outage would make every branch safe by construction and leave nothing for a contingency to violate. All ratings are then scaled so the intact base case peaks at 0.85 of the tightest branch's limit, which places the operating point at a stated fraction of the constraint instead of at whatever utilisation a file revision happens to encode. The network used in the paper has physical ratings on all but a single branch, so the substitution is nearly inert there and the calibration is a uniform rescaling.

Generation adequacy. Renewable hosts are pinned to whatever the weather gives them and are never re-dispatched upward; the remaining swing is shared pro rata over the headroom of the other units inside their ramping bands, and the slack bus absorbs the residual including losses. Active-power limits are relaxed by a stated factor so that the binding constraint in the ensemble is the transmission network rather than the generation stack, which is the regime this paper studies; the thermal and voltage limits that the safety claim rests on are left exactly as the file gives them.

Splits. Contingencies are ranked by base-case loading of the tripped branch and assigned to training, validation and testing cyclically by rank. A split by contiguous rank would hand the held-out set the least heavily loaded contingencies and let a low test error mean only that testing was easy; interleaving gives every split the same spread of severity.

Table 2: Throughput against the exact solver at operating-interval and screening batch sizes. The solver's cost is per case and does not vectorise; the operator's is dominated by fixed overhead at batch one. Means over repeated passes on 132 held-out scenarios.

Batch	Solver (ms)	Operator (ms)	Speedup
1	4.091	9.244	0.4426×
8	32.73	12.67	2.583×
32	130.9	22.13	5.917×
128	523.7	70.19	7.461×

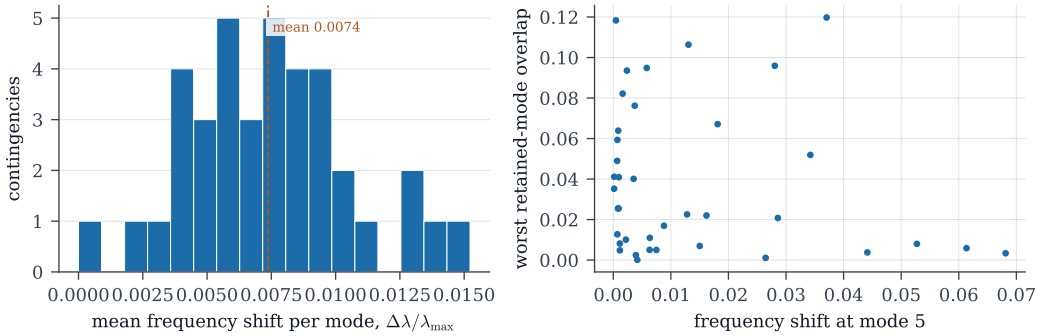


Figure 2: What a contingency does to the spectrum, over the $n = 38$ single-branch contingencies of the pool. The left panel is the distribution of the mean per-mode frequency drift relative to $\lambda_{\max}$ and the right panel is the worst retained-mode eigenvector overlap against the drift at a fixed mid-band mode. The two panels are the pair the retention-rule argument rests on, and they disagree, since the frequency of a mode barely moves while the direction of the mode moves enough that a filter evaluated at the right frequency multiplies different content.

C Architecture and training details

Each block applies the learned spectral or message-passing mixing to a normalised latent, updates every edge from its own embedding and both endpoint embeddings unless the variant is the topology-blind floor, and then applies a pointwise residual multilayer perceptron. Voltage magnitude, voltage angle and branch loading are read out by separate heads; the branch head additionally sees the raw edge features. Targets are normalised by fixed scales fitted on the training split only, and voltage angles are centred on the slack, which is the only angle the problem pins down. The branch term of the loss is masked to in-service branches, because asking the operator to reproduce a zero on a tripped branch teaches it that a tripped line and an unloaded one look alike, which is exactly the confusion a shield cannot afford. All architectures share the input featurisation, optimiser, learning-rate schedule, early-stopping rule and epoch budget, and the tables report parameter counts alongside errors so that a capacity explanation can be checked.

Two implementation details are load-bearing. The spectral basis for a topology is computed once and cached, which turns hundreds of thousands of eigen-decompositions into a few dozen and, more importantly, guarantees the basis does not change while a single action is being optimised. Evaluation is grouped by topology before batching, since a batch mixing two topologies would silently evaluate one of them in the other's eigenbasis and the resulting error would look like a modelling failure rather than a bookkeeping one.

D Spectrum movement, batching and runtime

The pair the argument rests on. Figure 2 plots the per-contingency frequency drift against the eigenvector rotation. The drift is small enough that the index and band rules are evaluated at nearly the same frequencies, and the rotation is not, which is why the retention rule matters less than the basis it is applied to. Both quantities are computed on the same normalized Laplacian the operator uses, with the same edge weights, so the measurement and the model see the same object.

Why the projection is affordable. The shield's cost per control interval is 100.2 ms on one CPU thread, against 4.091 ms for one exact solve, so it is 24.5 solves' worth of work. Almost all of that is

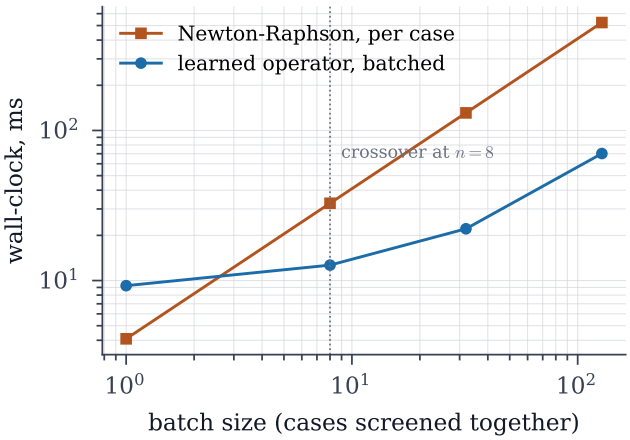


Figure 3: Wall-clock against batch size for the exact solver, which pays a fixed cost per case, and for the batched learned operator, on the same $n = 132$ held-out cases. Single queries favour the solver; the crossover sits between a batch of one and a batch of eight, and x-axis positions to the right of it are the regime in which a surrogate is worth its error.

Table 3: Safety on held-out topologies, judged by the Newton-Raphson solver rather than by the surrogate. Violation and islanding rates are fractions of control intervals; the false-safe rate is the fraction of shield-approved actions the solver then rejected, and is reported only for the shielded arm since the unshielded one has no approval step. Lower is better everywhere except intervals evaluated.

Controller	Violation	Islanded	False-safe	Cost ($)	Intervals
No control	0.8788	–	–	3760	132
PPO-Lagrangian	0.9133	0.1933	–	3379	150
PPO-Lagrangian + shield	0.8733	0.02	–	3940	150

operator passes rather than solver calls, since the enumeration is one pass over stacked bases and the projection's line search is one pass over trial step lengths, which is what the counts in Table 2 buy.

E Additional measurements

Per-seed tables, per-variant training wall-clock, the full per-contingency drift table and the shield's projection-step histogram are written by the pipeline into the run directory alongside the numbers used in the body.

L2RPN cross-check. The environment used in this paper is a self-contained one built on our own verified solver, which is what allows the solver to be checked against an independent implementation and the constraint bookkeeping to be controlled exactly. The released code also contains an adapter that runs the same kind of sequential decision problem inside Grid2Op and the L2RPN case-14 sandbox, so the operating regime can be compared against the established benchmark; those numbers are reported separately from the ones above because they use a different power-flow engine and a different constraint bookkeeping, and mixing the two would make the verification claim meaningless.

The cross-check was executed on a 14-substation, 20-line environment over 96 five-minute intervals under a do-nothing policy, which is the comparison that decides whether the two regimes can be mentioned in the same sentence. It cannot, since a do-nothing policy overloads no line across the 96 intervals and disconnects none, with a mean worst loading of 0.8356 and a worst single value of 0.9297. The sandbox at its default setpoint therefore sits below its own limits, while the ensemble this paper studies is drawn at the boundary where 0.8788 of states already violate something. What the cross-check establishes is a scope statement and not a safety result, because the regime where a projection shield has anything to do is a stressed one while the standard benchmark's default operating point is not. We report that rather than tuning the sandbox until both regimes agree.

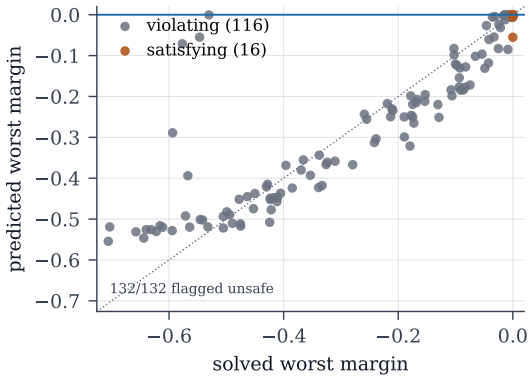


Figure 4: The feasibility verdict of the band operator on held-out contingencies, predicted worst constraint margin against the margin the solver measured ($n = 132$ scenarios; the reference is the verified Newton-Raphson solver and the dashed line is the approval threshold). The failure is one-sided, since every state, including the 0.1212 of them that satisfy every limit, is predicted infeasible, and no action is approved on the strength of the surrogate's word.

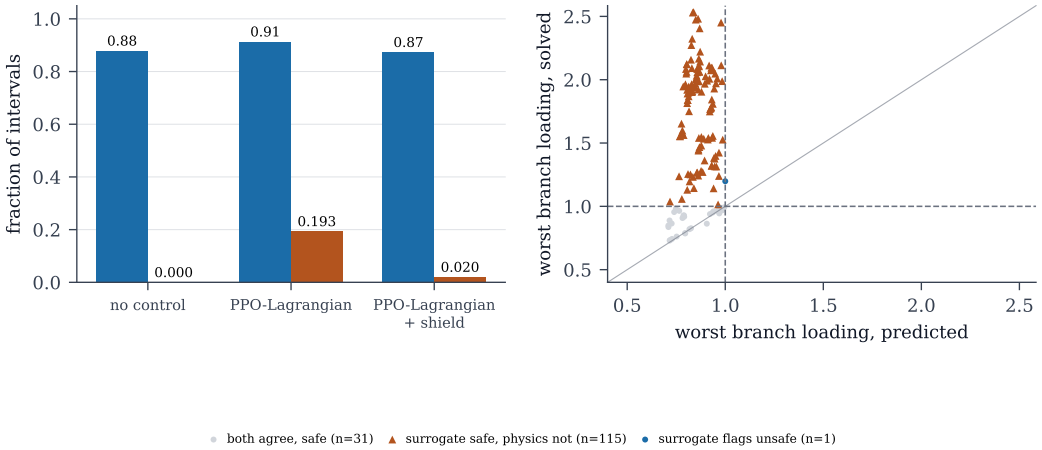


Figure 5: Safety on held-out topologies as the solver judged it and never the surrogate. The left panel gives constraint-violation and islanding rates over $n = 132$ intervals per arm with the do-nothing dispatch as the baseline (left), and the worst branch loading the band operator predicted against the one the solver measured, for every interval of the shielded arm (right). The projected actions still violate a limit in 0.8733 of intervals while the islanding rate falls from 0.1933 to 0.02, which is where the shield's contribution is measurable and where the surrogate's error lands.

Availability of code and data

Every number reported here is read from the committed per-run result records by `gridpino.reporting`, which emits the manuscript's macros, and by `tools/check_macros.py`, which fails loudly when a macro the manuscript uses is absent from the run, so a stale figure or a retyped number cannot ship unnoticed. The pipeline, the unit tests, the Kaggle kernel and the Modal app that run the same study on an accelerator are available from the author on request. The networks are MATPOWER and pandapower cases, the cross-check uses the public Grid2Op L2RPN case-14 sandbox, and no non-public data is used.

Use of AI tools

Generative AI tools were used to implement the experiments and to draft this manuscript, and the prose was then revised by hand. Their output was checked by re-execution rather than by reading, since every numerical claim is emitted as a LaTeX macro from the run's JSON rather than transcribed, the solver is verified against pandapower over 59 states, the spectral basis, the capacity matching of the two retention rules, the topology-disjointness of the splits and the monotonicity of the projection are each covered by unit tests, and the references were checked against their sources. The authors directed the work and take responsibility for all content.